

- exhaustive { all solutions }

Example 1: Input: nums = [1, 2, 3]

Output: $[[1], [2], [3], [1, 2], [1, 3], [2, 3], [1, 2, 3], []]$

Example 2: Input: nums = [0]

Output: $[[1], [0]]$

 $[1, 2, 3]$

DAS

Don't choose
the number

Choose the
number

$$Res = []$$
$$\text{Sol} = []$$
$$Res = [1] \quad Reo = [1, 3]$$
$$S_0 = [3]$$

First ⁶Temporarily $\rightarrow$ then pop from Sol
append [3] to Sol. and continue

class Solution:

def subsets(self, nums: List[int]) -> List[List[int]]

n = len(nums)

res, sol = [], []

def backtrack(p):

if p == n:

res.append(sol[:])

return

Don't pick nums[p]

backtrack(p+1)

Pick nums[p]

sol.append(nums[p])

backtrack(p+1)

sol.pop()

backtrack(0)

return res

Time: $O(2^n)$

Space: $O(n)$